Early TEFCA Participation Was Not Associated With Lower Hospital Mortality

A Cross-sectional analysis of 691 U.S. hospitals

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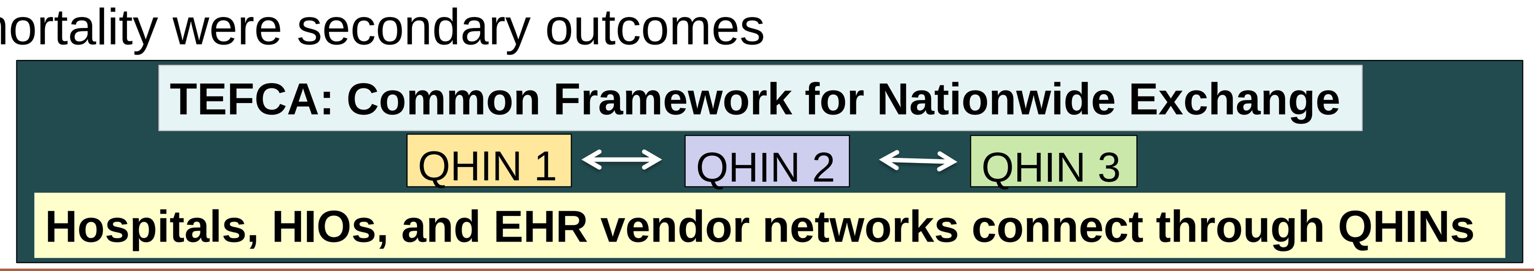
Code, methods, and materials

Background & Objective

The Trusted Exchange Framework and Common Agreement (TEFCA) is a national framework for electronic health information exchange (HIE). It connects separate networks through Qualified Health Information Networks (QHINs).[1]

Evidence gap. Prior HIE research has reported operational and utilization benefits, but evidence of lower mortality remains limited and largely observational.[2–11] TEFCA's relationship with mortality during early implementation is unclear.

Objective. Determine whether U.S. non-federal acute care hospitals participating in TEFCA, versus hospitals planning to participate, were associated with lower mortality. The primary outcome was CMS Hybrid Hospital-Wide Mortality (Hybrid HWM); heart failure and pneumonia mortality were secondary outcomes



Methods

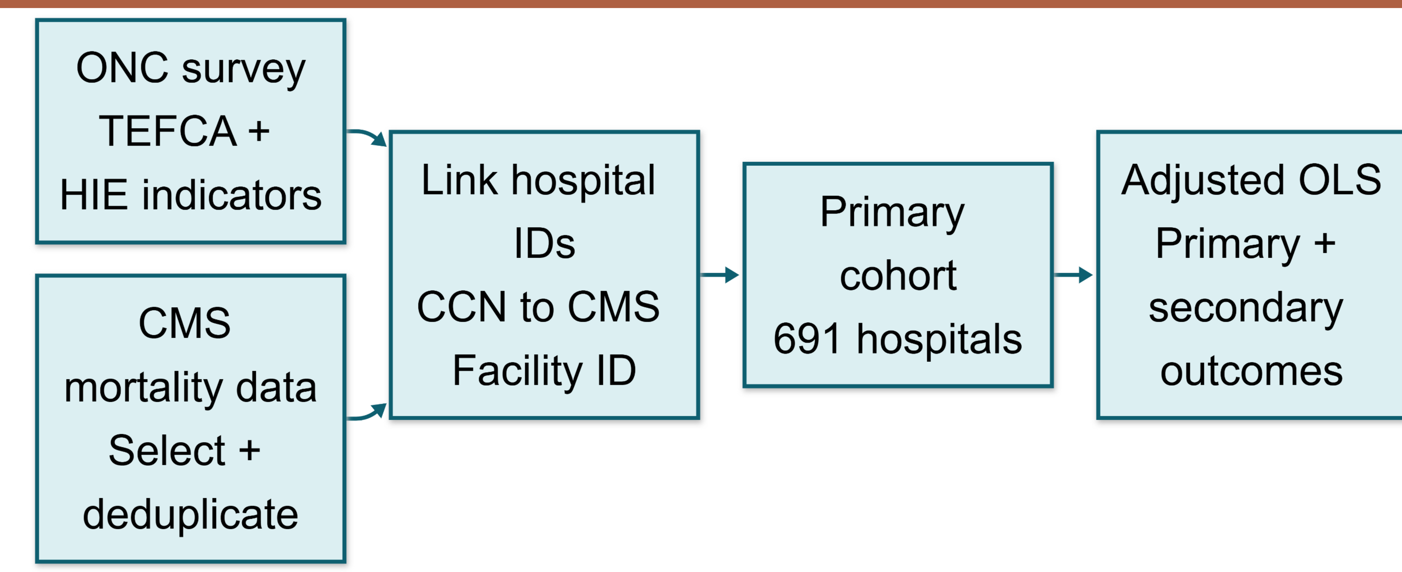


Figure 1. Data linkage and analytic workflow.

Hospital-level ONC survey data (2023–2024)[12] were linked to CMS Hybrid HWM data (7/1/2023–6/30/2024) and heart failure/pneumonia mortality data (7/1/2021–6/30/2024).[13] Adjusted OLS models compared current and neither status with planned participation, controlling for national network, EHR vendor network, and HIO participation, log CMS denominator (total eligible patient population), state, and survey year. *Note.* Missing TEFCA and other network participation responses were coded as nonparticipation.

Results

Primary analysis.

691 hospitals

- 112 (16.2%) currently in TEFCA
- 289 (41.8%) planned TEFCA
- 290 (42.0%) neither

In the adjusted primary model: current participation associated with 0.108-point higher Hybrid HWM score than planned participation (95% CI, 0.001–0.216; *p* = .047).

Robustness and secondary outcomes. With HC3 robust standard errors, the 95% confidence interval included zero (-0.010 to 0.227; *p* = .072). Current-versus-planned estimates also not significant for heart failure (β = −0.024; *p* = .927) or pneumonia mortality (β = 0.479; *p* = .114). Participation in EHR vendor networks and larger CMS denominators associated with lower Hybrid HWM scores.

Key finding: The analyses did not provide robust evidence that early current TEFCA participation was associated with lower hospital mortality than planned participation.

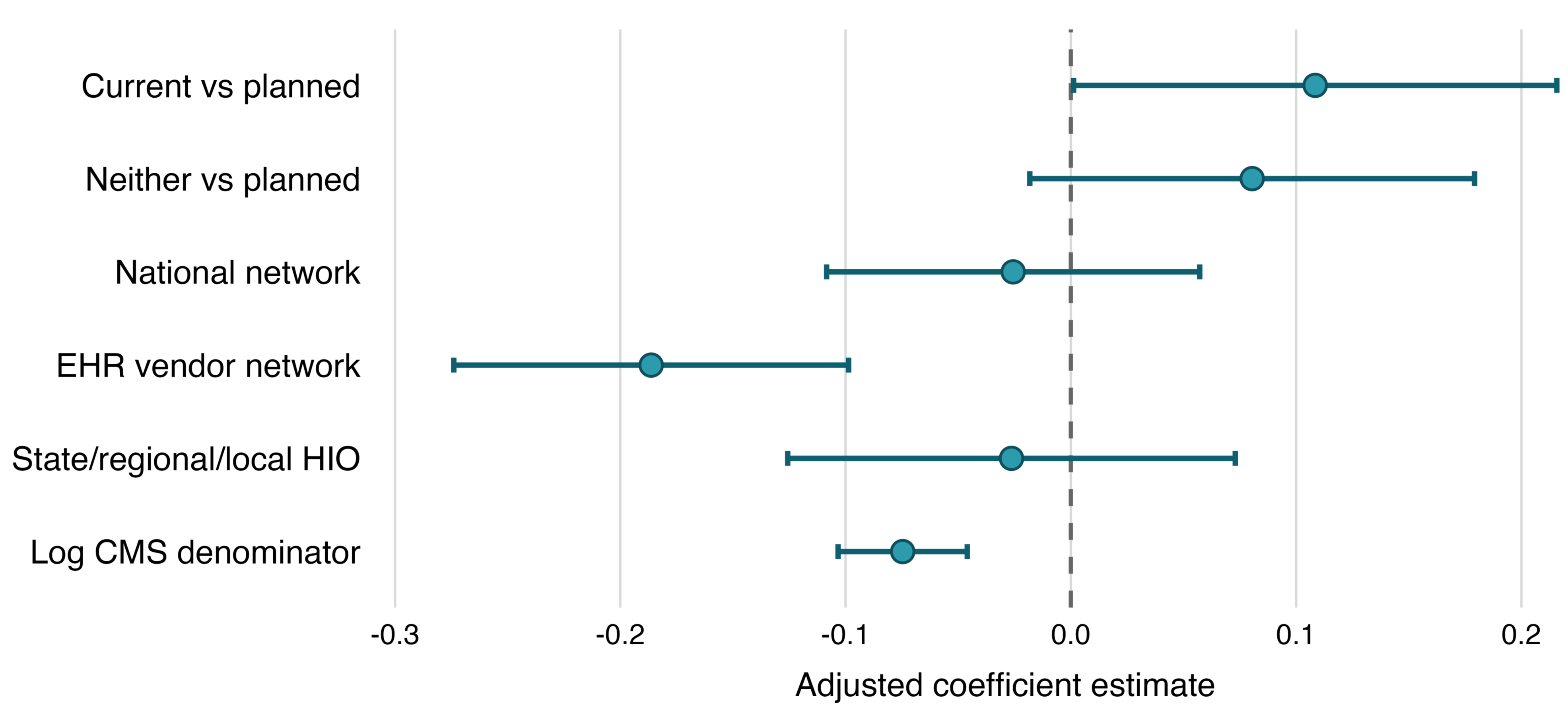
Tables & Charts

Table 1. TEFCA Participation Estimates Across Mortality Analyses

Analysis: current vs planned TEFCA hospitals	β	95% CI	P-value
Primary Hybrid HWM model	0.108	0.001 to 0.216	.047
Primary model, HC3 robust SEs	0.108	−0.010 to 0.227	.072
Heart failure mortality	−0.024	−0.545 to 0.497	.927
Pneumonia mortality	0.479	−0.115 to 1.073	.114

Note. Positive coefficients indicate higher mortality values; planned TEFCA participation was the reference category.

Figure 2. Adjusted associations with Hybrid HWM



Note. Points represent adjusted OLS coefficients; error bars represent 95% confidence intervals. Planned TEFCA participation was the reference category.

Interpretations & Limitations

Interpretation. Early TEFCA participation was not robustly associated with lower mortality than planned participation. The small positive estimate was sensitive to the standard-error method and absent for secondary outcomes; it should not be interpreted as evidence of harm. Effects may require longer, fuller adoption and routine exchange use.

EHR vendor network participation was associated with lower Hybrid HWM, possibly reflecting more established infrastructure or workflow integration, but this secondary finding does not establish causality.

Limitations.

- Cross-sectional hospital comparisons cannot establish causality or within-hospital change.
- Surveys measured reported participation, not HIE volume, quality, or workflow integration.
- Planned hospitals may already use other networks, narrowing the contrast.
- Unmeasured hospital characteristics may confound estimates.
- The primary outcome window was not fully post-TEFCA; secondary windows largely predated implementation.

Future work. Longitudinal studies should evaluate sustained TEFCA use, actual exchange activity, and later clinical outcomes (e.g., difference-in-differences).

Conclusion

In this early hospital-level analysis, current TEFCA participation was not robustly associated with lower mortality than planned participation, and the small positive Hybrid HWM estimate was not reproduced across secondary outcomes. Because planned hospitals could already use other exchange networks, this was not a comparison with no electronic exchange. EHR vendor-network participation was associated with lower Hybrid HWM, but this observational finding does not establish causality. Longitudinal studies should evaluate sustained TEFCA use and actual exchange activity as implementation matures.

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